

DATABASE SYSTEMS — LAB 06

Course: Database Systems

Topic: ER Modeling & Concepts

◇ PART A — Theory Questions

Q1.

A) Explain database models according to the degree of abstraction.

B) What is meant by an instance of an entity? Give an example.

Q2.

What is an entity? Also explain:

- What should be considered an entity
- What should NOT be considered an entity

Q3.

What are attributes? Explain different types with examples.

Q4.

A) What is a relationship in ERD? Explain types with examples.

B) How are multi-valued attributes handled in a relational database? Explain with the help of an example

◇ PART B — Short Questions

Q5.

A hospital system maintains:

- Patient_ID
- Full_Name (First + Last)

- Date_of_Birth
- Age
- Contact_Number (a patient may have more than one)

Identify:

- Entity
- Attributes
- Composite attribute
- Multi-valued attribute
- Derived attribute
- Which attribute should NOT be stored
- Reason

Q6.

A system stores:

- Employee_ID
- Name
- Salary
- Bonus (calculated as 10% of salary)

Identify:

- Entity
- Attributes
- Derived attribute
- Which attribute should NOT be stored
- Give reason

Q7.

A student system stores:

- Student_ID
- Name
- Email
- A student can have multiple phone numbers

Identify:

- Entity

- Attributes
- Primary key
- Multi-valued attribute

◇ PART C — ER Diagram

Use rectangles for entities, ovals for attributes, and underline primary keys.

Q8.

A school system stores:

- Students (Student_ID, Name, Age)
- Teachers (Teacher_ID, Name, Subject)
- A teacher teaches many students
- A student is taught by one teacher

Task:

Draw the ER diagram with proper notation.

Q9.

An online shopping system stores:

- Customers (Customer_ID, Name)
- Products (Product_ID, Product_Name, Price)
- A customer can buy multiple products
- A product can be bought by multiple customers
- Each purchase has Purchase_Date

Task:

Draw the ER diagram with proper notation.

Q10.

An online shopping system stores:

- Customers (Customer_ID, Name, Address, Phone_Number)
- Products (Product_ID, Product_Name, Price)
- A customer can buy products

Task:

Identify the entities and attributes clearly, and draw the ER diagram.

◇ PART D — IDENTIFY

Q11.

Library system stores:

Books (Book_ID, Title)

Authors (Author_ID, Name)

A book can have multiple authors

An author can write multiple books

Task:

Relationship type?

Do we need a new entity?

What will be its attributes?

Q12. Car Rental System

A system stores:

Customers (Customer_ID, Name)

Cars (Car_ID, Model)

A customer can rent many cars

A car can be rented by many customers over time

Each rental has Start_Date and End_Date

Identify:

Relationship type

Important attributes

Any derived attribute?